Report 1: Are Headtracking Measures an Indicator of Depressive Symptoms?

Preliminary Data Visualization, Descriptive Statistics, and Inferential Analysis

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1 Introduction

Depression is a leading cause of disability worldwide, yet objective behavioral markers remain limited. While self-report instruments such as the Patient Health Questionnaire (PHQ-9; Kroenke et al. 2001) are widely used for screening, they are inherently subjective. An emerging line of research investigates *psychomotor retardation*—the slowing of physical movement and thought—as an observable correlate of depression (Bennabi et al., 2013). The anhedonia component of depression may manifest as reduced exploratory behavior in stimulating environments.

Recent work by Srivastava et al. (2025) demonstrated that headtracking data collected during 360° virtual reality (VR) video viewing could distinguish individuals based on depression severity. Using ANCOVA to control for anxiety (GAD-7 and STAI-T), they found that *scanning speed*—the rate of angular head movement—was significantly lower in the moderate-severe depression group ($p < .001$, $\eta^2 = .295$, Cohen's $d = 1.64$). The present study is a **replication** of this work, conducted with college students in India using a Meta Quest 3 VR headset.

Participants viewed five 360° videos spanning different emotional contexts and completed standardized measures of depression (PHQ-9), anxiety (GAD-7; Spitzer et al. 2006; STAI-T; Spielberger et al. 1983), mood (PANAS-SF; Watson et al. 1988), presence, and simulator sickness (VRNQ-VRISE; Kourtesis et al. 2019). Since depression and anxiety are highly covariant (Kalin, 2020), this overlap must be accounted for. This report presents preliminary descriptive statistics, visualizations and initial inferential tests to characterize the dataset before more advanced analyses in Report 2.

2 Methods

2.1 Participants

Forty college students (36 male, 4 female; age $M = 22.8$, $SD = 1.8$, range 19–27) participated. Sixteen (40%) reported prior VR experience. Participants were recruited via convenience sampling.

2.2 Materials

Five 360° videos were used:

- **V1**: Abandoned Buildings (60s) — eerie, low valence
- **V2**: Evening at the Beach (60s) — pleasant, calm
- **V3**: Campus (60s) — neutral, familiar
- **V4**: The Nun Horror (≈3 min) — high arousal, negative
- **V5**: Tahiti Surf (≈3 min) — high arousal, positive

Videos were viewed on a Meta Quest 3 headset.

2.3 Measures

Headtracking: Pitch (X), Yaw (Y) and Roll (Z) rotation changes and speeds were recorded. Summary measures per participant per video included mean rotation speed, SD, median speed, mean absolute rotation change, angular range, and peak speed. **Clinical scales**: PHQ-9 (depression, 0–27; cutoff ≥ 10; Kroenke et al. 2001), GAD-7 (anxiety, 0–21; Spitzer et al. 2006), STAI-T (trait anxiety, 20–80; Spielberger et al. 1983). **Affect and experience**: PANAS-SF (positive/negative affect pre- and post-VR; Watson et al. 1988), VRNQ-VRISE (simulator sickness, 5–35; Kourtesis et al. 2019), valence and arousal ratings (1–9 per video), and a 5-item presence measure (5–35 per video).

2.4 Depression Group Classification

Participants were classified as *Depressed* (PHQ-9 ≥ 10; $n = 8$) or *Non-Depressed* (PHQ-9 < 10; $n = 32$) using the established clinical cutoff for moderate depression (Kroenke et al., 2001). The original study by Srivastava et al. (2025) used three depression groups (minimal, mild, moderate-severe) with ≥ 10 marking the moderate-severe threshold; we adopt the same cutoff for a binary split given our smaller sample.

2.5 Statistical Approach

Descriptive statistics (means, SDs, medians, ranges) and Shapiro-Wilk normality tests were computed for all key variables. Pearson correlations assessed bivariate relationships. Welch's independent-samples t-tests compared headtracking measures across depression groups (robust to moderate non-normality with $N = 40$; Cohen's d for effect sizes). Bonferroni correction was applied to the 25 simultaneous t-tests ($\alpha_{\text{corrected}} = 0.05/25 = 0.002$). Paired t-tests assessed pre-post PANAS changes. Outliers were detected via IQR (clinical scores) and z-score ($|z| > 3$; headtracking). All analyses used Python 3.12 (`pandas`, `scipy`, `seaborn`, `matplotlib`); $\alpha = 0.05$ unless noted.

3 Results

3.1 Demographics and Sample Characteristics

The sample was predominantly male (90%) with a narrow age range (19–27 years). Figure 1 shows the demographic distributions.

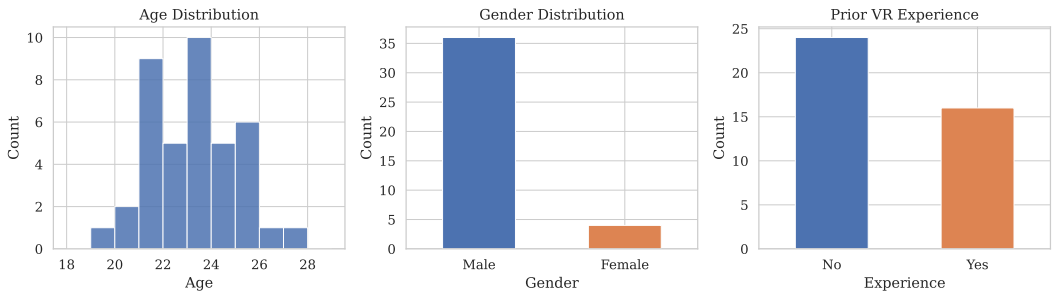


Figure 1: Distributions of age, gender and prior VR experience across the 40 participants.

3.2 Clinical Measures

PHQ-9 scores ranged from 0 to 18 ($M = 6.03$, $SD = 4.63$, $Mdn = 4.5$), with 50% of participants in the minimal range (0–4), 30% mild (5–9), 12.5% moderate (10–14), and 7.5% moderately severe (15–19). Eight participants (20%) met the threshold for moderate depression (PHQ-9 ≥ 10). GAD-7 scores ranged from 0 to 18 ($M = 5.00$, $SD = 4.31$). STAI-T scores ranged from 21 to 73 ($M = 45.00$, $SD = 14.63$).

Shapiro-Wilk tests indicated that both PHQ-9 ($W = 0.897$, $p = .002$) and GAD-7 ($W = 0.883$, $p < .001$) distributions deviated significantly from normality, with a positive skew reflecting the predominantly non-clinical sample. STAI-T was marginally non-normal ($W = 0.944$, $p = .045$).

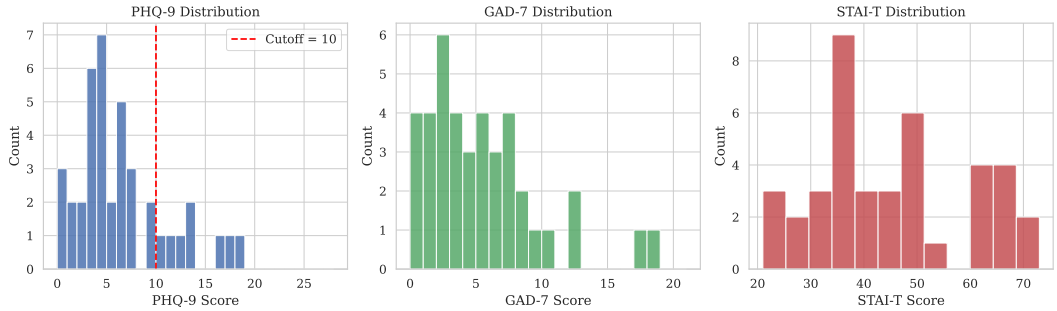
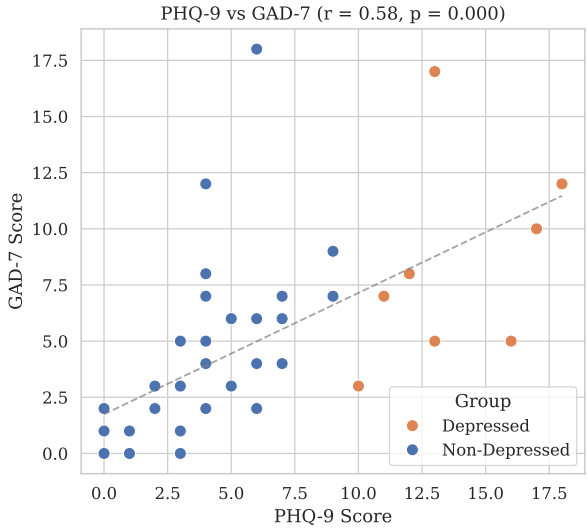


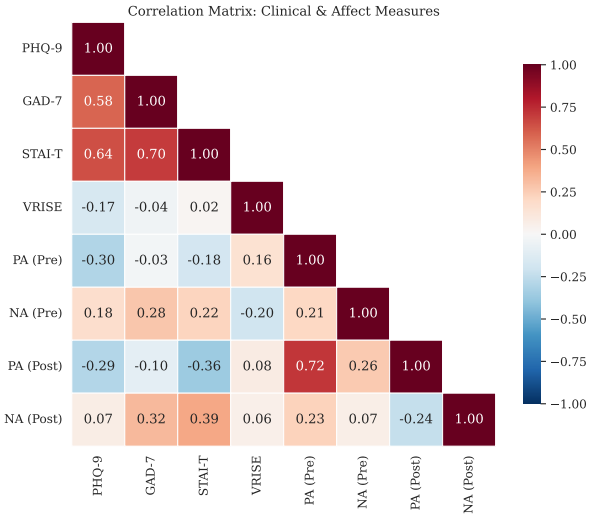
Figure 2: Distributions of PHQ-9, GAD-7 and STAI-T scores. Dashed red line on PHQ-9 indicates the clinical cutoff (≥ 10).

3.3 Depression–Anxiety Covariance

PHQ-9 and GAD-7 were strongly correlated ($r = .579$, $p < .001$), consistent with documented comorbidity (Kalin, 2020). PHQ-9 and STAI-T were also strongly correlated ($r = .642$, $p < .001$), underscoring the need to control for anxiety when isolating depression-specific effects.



(a) PHQ-9 vs. GAD-7 scatter plot.



(b) Correlation heatmap of clinical and affect measures.

Figure 3: (a) Scatterplot showing the strong positive relationship between PHQ-9 and GAD-7 scores. (b) Correlation matrix showing inter-relationships among all clinical and affect measures.

3.4 Video-Level Emotional and Presence Ratings

The five videos elicited distinct emotional profiles (Table 1). V5 (Tahiti Surf) had the highest valence and arousal, while V4 (Horror) had the lowest valence but high arousal. V1 scored

highest on presence; V4 lowest.

Table 1: Mean (SD) valence, arousal and presence scores by video.

Video	**Valence**	**Arousal**	**Presence**
V1: Abandoned Buildings	5.20 (1.67)	4.60 (2.22)	29.60 (4.41)
V2: Beach	6.53 (1.75)	4.65 (2.03)	27.25 (5.31)
V3: Campus	6.15 (1.61)	4.33 (2.16)	26.95 (4.52)
V4: Horror (Nun)	3.67 (1.85)	5.95 (1.68)	19.85 (5.44)
V5: Tahiti Surf	7.12 (1.68)	6.50 (1.69)	28.02 (4.79)

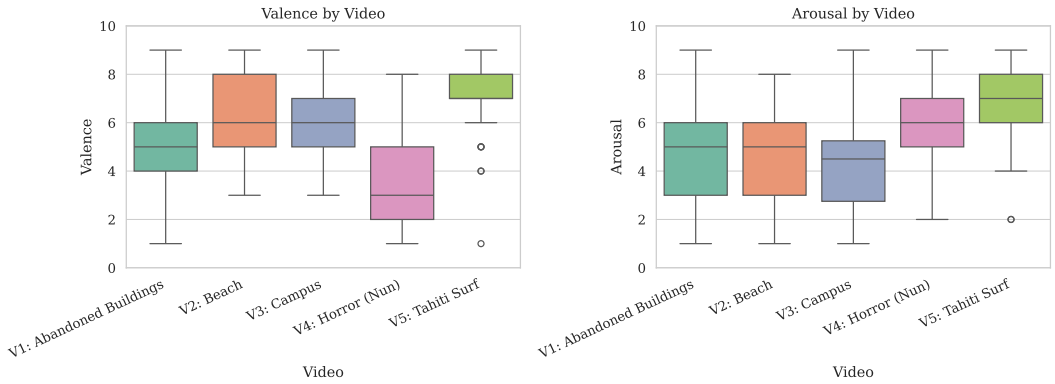


Figure 4: Box plots of valence and arousal ratings across the five videos.

3.5 Headtracking Measures

Mean rotation speed varied across videos: V1 (Abandoned Buildings) was highest (M = 39.08°/s), consistent with its exploratory nature, while V4 (Horror) was lowest (M = 24.17°/s). Formal ANOVA testing is deferred to Report 2.

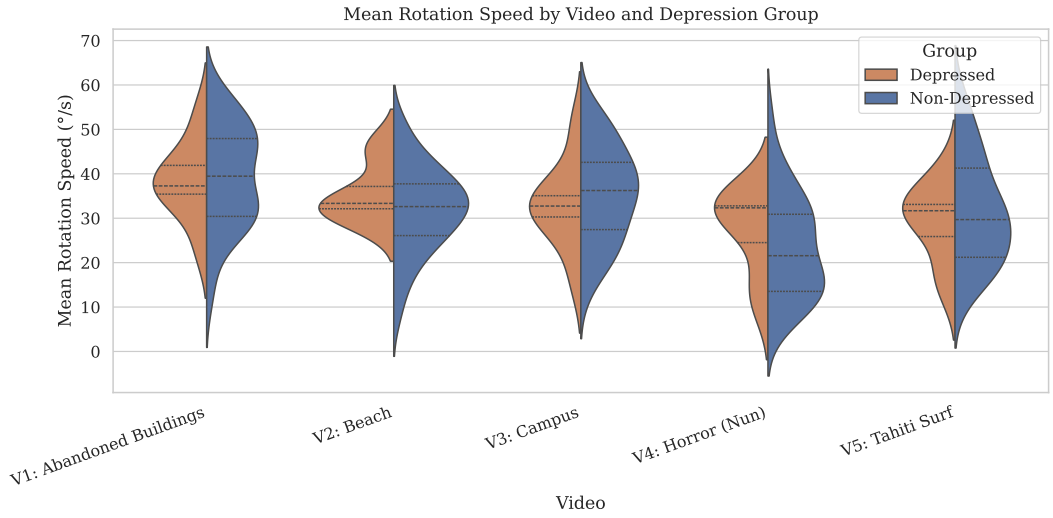


Figure 5: Split violin plots of mean rotation speed by video and depression group (Non-Depressed: blue; Depressed: orange).

3.6 Headtracking by Depression Group

Welch's t-tests yielded largely non-significant results (Table 2). Only V4 angular range reached uncorrected significance ($t = 2.21$, $p = .048$, $d = 0.83$), with the depressed group exploring *greater* range—counter to the psychomotor retardation hypothesis. After Bonferroni correction ($\alpha_{\text{corrected}} = .002$), **no comparisons remained significant**. Pearson correlations between PHQ-9 and participant-averaged headtracking were near zero: mean total speed ($r = -.060$, $p = .712$), yaw speed ($r = -.081$, $p = .617$), SD of total speed ($r = -.158$, $p = .329$).

Table 2: Selected Welch's t-test results for mean total speed and total angular range by depression group per video. Cohen's d quantifies effect size. No comparison survived Bonferroni correction ($\alpha_{\text{corrected}} = .002$).

Video	**Mean Speed (Total)**			**Total Angular Range**		
	t	p	d	t	p	d
V1: Abandoned Buildings	−0.24	.816	−0.08	−0.22	.831	−0.07
V2: Beach	1.22	.239	0.38	1.76	.104	0.65
V3: Campus	−0.64	.532	−0.24	0.16	.878	0.05
V4: Horror (Nun)	1.07	.303	0.37	2.21	.048	0.83
V5: Tahiti Surf	−0.66	.523	−0.21	−0.36	.727	−0.16

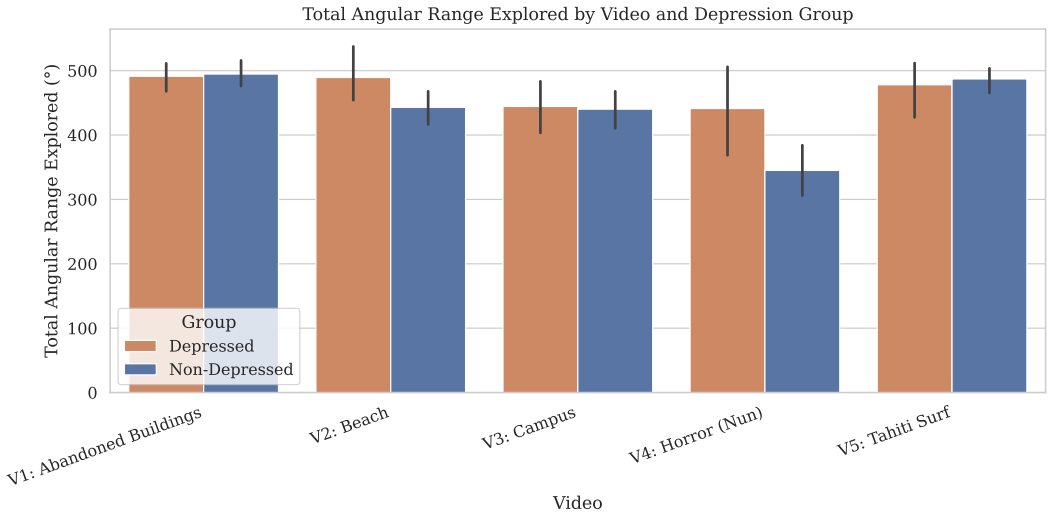


Figure 6: Total angular range explored by video and depression group. Error bars represent 95% CIs.

3.7 PANAS: Pre- and Post-VR Mood

Positive Affect decreased significantly from pre-VR ($M = 34.52$) to post-VR ($M = 32.15$; paired $t(39) = 2.09$, $p = .043$, Cohen's $d = -0.33$). Negative Affect showed a non-significant increase ($M = 12.70 \rightarrow 13.95$; $t(39) = -1.06$, $p = .294$, $d = 0.17$).

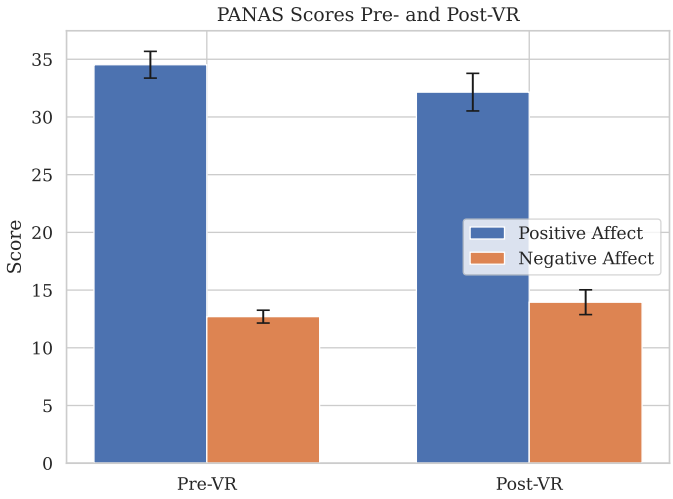


Figure 7: PANAS positive and negative affect scores before and after VR exposure. Error bars represent standard errors.

3.8 Outlier Analysis

Using the IQR method, 3 PHQ-9 and 2 GAD-7 outliers were identified but retained as genuine clinical variability. No headtracking outliers exceeded $|z| > 3$. One participant with a low VRISE score (17) was flagged for sensitivity analysis.

4 Conclusion and Future Directions

4.1 Summary of Findings

1. The sample is predominantly non-clinical (80% below PHQ-9 cutoff), with only 8 in the depressed group.
2. Depression and anxiety are strongly correlated ($r = .58$), confirming the need to control for this covariance.
3. Headtracking speed varies descriptively across videos (V1: 39.08°/s; V4: 24.17°/s); formal ANOVA deferred to Report 2.
4. Welch's t-tests showed **no significant differences** in headtracking speed by depression group after Bonferroni correction. The original finding (Srivastava et al., 2025) was **not replicated**, likely due to limited power.
5. One uncorrected effect (V4 angular range, $p = .048$, $d = 0.83$) was in the opposite direction and likely a Type I error.
6. Positive affect decreased after VR exposure ($p = .043$, $d = -0.33$).

4.2 Limitations

The key limitation is the small and unbalanced sample ($n_{\text{dep}} = 8$ vs. $n_{\text{nondep}} = 32$), which severely limits statistical power. The predominantly male sample (90%) limits generalizability. Additionally, the current binary classification does not exploit the full range of PHQ-9 scores.

4.3 Plans for Report 2

Report 2 will employ PHQ-9 as a continuous predictor (regression/GLM) and include GAD-7 and STAI-T as covariates (ANCOVA) to disentangle depression from anxiety effects. Mixed-effects models will account for the within-subject structure across videos, and video-type interactions will be tested. Additional analyses will include mediation/moderation with presence and sickness scores, power analysis, alternative multiple-comparison corrections (Holm, Benjamini-Hochberg), and one-way ANOVA for cross-video speed differences.

5 Author Contributions

The complete code and analysis pipeline are available at: `https://github.com/parv-the-coder/BRSM_Project`

Parv Shah (2025201093): Developed the data processing pipeline, implemented the statistical analyses (descriptive statistics, inferential tests, correlation analyses, outlier detection), produced the visualizations, interpreted the results and drafted the report.

Hardik (2025201046): Conducted the literature review, performed exploratory data analysis to inform the statistical approach and co-edited the report.

Gaurav (2025201065): Conducted background research on headtracking methodologies, data cleaning and verification and co-edited the report.

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